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Quoodle: A privacy-first, zero-configuration quiz-distribution tool to support self-regulated learning

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Software

- Review
- Repository
- Archive

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Summary

`Quoodle` is a minimalist, self-contained web application for creating and distributing multiple-choice knowledge checks in educational settings. The workflow is familiar to anyone who used a doodle.com to schedule a meeting (see fig. 1 below). A teacher uploads a spreadsheet of questions; quoodle returns a shareable URL and QR code for learners, along with a separate URL through which the teacher can view aggregated results. Learners can answer the quiz on any device, receive immediate per-question feedback with an explanation, and see their score and basic information on upon completion. The teacher sees class-level statistics — success rates per question, distribution of selected distractors, average completion time, and a tab-switching indicator — but no individual submission records.

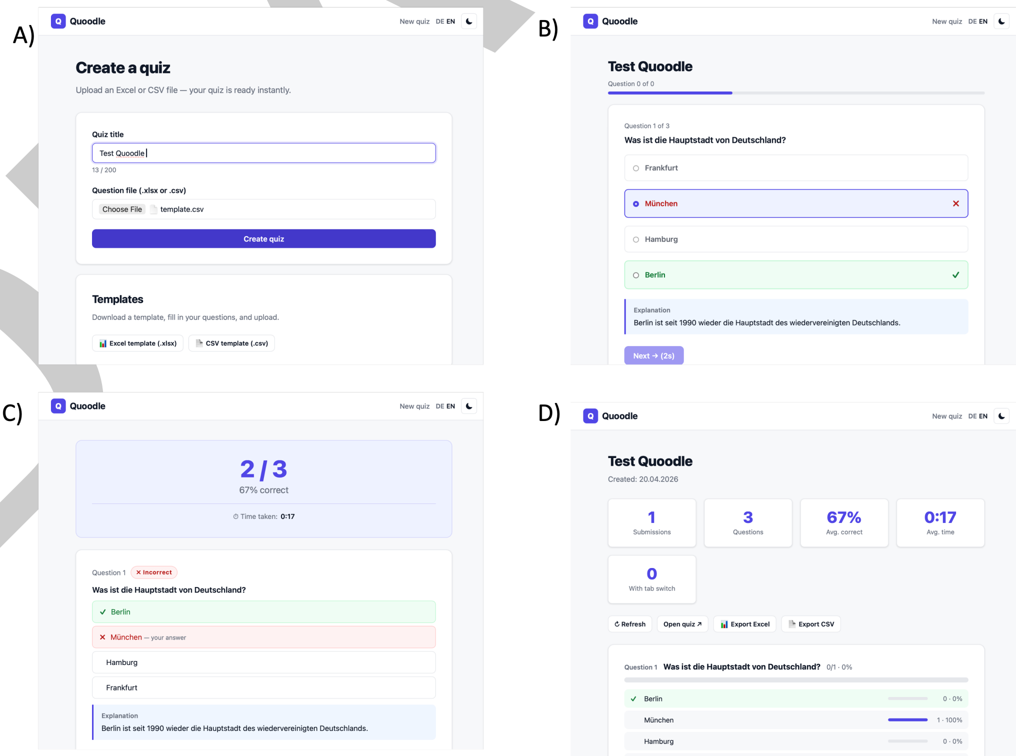

Figure 1: Figure 1. Overview of Quoodle. A.) Landing page to create a quizz. B.) Quizz-taking view for students with feedback. C.) Quizz-result view for students. D.) Result-view for teachers.

Statement of need

Formative assessment through short knowledge checks is a well-established classroom practice that is highly efficient in improving student learning (Hattie & Timperley, 2007). However, in Germany less than 20% of university students report receiving regular feedback (Multrus et al., 2017). Many tools exist to close this gap from proprietary platforms such as Kahoot! and Quizlet to open-source alternatives like ClassQuiz (Mawoka, 2022). Existing tools, however, impose one or more barriers to their wide-spread adoption of quizzes to support student learning.

1. **Student accounts.** Most platforms require learners to create an account or join a session identified by a persistent pseudonym.
2. **Third-party data processing.** Hosted platforms typically process student interaction data on servers outside the educator's legal jurisdiction, creating friction with the EU General Data Protection Regulation (European Parliament and Council, 2016) and regional school-data-protection directives.
3. **Infrastructure complexity.** Open-source alternatives often require Docker, a separate database server, a message broker, and a reverse proxy — a barrier for individual teachers, small schools, and resource-constrained institutions.
4. **Tracking and telemetry.** Even where not strictly required, many tools contact analytics services, CDNs, or sentry endpoints on every page load, surfacing data outside the trust boundary the learner expects.
5. **Correctness-only feedback without explanations.** Most existing tools are not designed to foster self-regulated learning only provide feedback on the correctness or incorrectness of the responses without offering explanations.

`Quoodle` addresses these barriers by focusing on a very limited application scenario rather than by configuration. Even for those teachers with access to learning-management systems, quoodle is faster, easier and simpler than most LMS-based tools that do not allow the import of existing multiple-choice tests.

- **No student or teacher accounts.** A quoodle is a transient activity to improve self-regulated learning. Learners access the quiz through a URL and are neither asked for a name nor tracked across sessions.
- **No personally identifiable data is ever stored.** The database contains only aggregate counters — the number of submissions per quiz, how often each question was answered correctly, and how often each answer option was selected. Individual submissions are not persisted.
- **Spreadsheet based Multiple-Choice Tests** Even though complex specifications for tests have been proposed and implemented in LMS [https://docs.moodle.org/501/en/Import_ these have a steep learning curve for teachers who prefer simple spreadsheet-based methods to store and share multiple-choice tests.
- **No external network requests.** All assets (stylesheets, scripts, fonts, QR codes) are served from the same origin as the application. QR codes are generated server-side in PHP, eliminating dependence on external QR services.
- **Minimal hosting requirements.** The only requirements for the hosting are PHP 8 with the `pdo_sqlite` and `zip` extensions — both present by default on essentially every commercial hosting provider. No database server, no container runtime, no build tooling, and no Node.js is needed.

`Quoodle` is intentionally limited to asynchronous, self-paced knowledge checks. It does not offer live, synchronous quiz sessions with scoreboards for individual learners; educators seeking that functionality are better served by tools such as ClassQuiz (Mawoka, 2022) or comparable live-quiz platforms.

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Pedagogical design

The student-facing interface implements immediate formative feedback, a practice with substantial support in the educational literature (Hattie & Timperley, 2007). Furthermore, how students perceive of the assessment matters to their outcome (Brown et al., 2009). When a learner selects an answer:

- The correct answer is highlighted in green and the selected answer, if different, in red.
- Any explanation provided by the educator (an optional spreadsheet column) is revealed inline.
- After a correct selection, the learner may advance immediately.
- After an incorrect selection, the "next" button is disabled for five seconds, forcing the learner to engage with the explanation before moving on. This follows the retrieval-practice-with-feedback design principle (Roediger & Butler, 2011) that immediate, elaborative feedback on incorrect responses produces larger learning gains than delayed or correctness-only feedback.

For educators, the statistics dashboard surfaces three categories of insight. These are signals for the educator, not surveillance of individuals; the underlying per-submission values are never persisted.

1. **Overall performance and engagement** - The number of learners who took the test, the average number of correct responses, the average time on task and the average number of tab-switches.
2. **Individual question difficulty** — The percentage of learners who answered each question correctly, color-coded for rapid scanning (green □ 75%, amber 50–74%, red $< 50\%$).
3. **Individual distractor analysis** — how often each wrong option was selected, useful for identifying misconceptions that a particular distractor captures.

Implementation

`Quoodle` consists of approximately 30 files organised into page-level scripts (`index.php`, `upload.php`, `quiz.php`, `submit.php`, `share.php`,`stats.php`, `export.php`), a library directory (SQLite access, QR code generator, XLSX reader and writer, internationalisation), and a lightweight JavaScript/CSS front-end with no build step and no framework dependency.

The QR-code generator implements the full ISO/IEC 18004 encoding pipeline (International Organization for Standardization, 2015). The XLSX reader and writer use PHP's built-in `ZipArchive` and `SimpleXML` extensions to parse and produce Office Open XML documents directly, without any external library. The writer supports multiple sheets, bold header rows, and shared-string deduplication.

Concurrent submissions are handled via SQLite's Write-Ahead Logging journal mode and `BEGIN IMMEDIATE` transactions, with an exponential-backoff retry loop bounded at ten attempts. Submissions that fail to record (e.g. on sustained lock contention) do not block the learner's feedback page — a deliberate trade-off that favours learner experience over perfect aggregation consistency in pathological cases.

The source code is availible at https://github.com/gerhi/quoodle. A hosted version can be accessed at http://www.gerrithirschfeld.de/quoodle.

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